

Chapter 16.8: The Divergence Theorem

V3: Generalizations of the FTC. I can state and apply Green's Theorem, Stokes' Theorem and the Divergence Theorem to solve problems in two and three dimensions. I can choose which theorem is appropriate for different integrals. I can compute curl and divergence of vector fields.

Mechanics

1. Consider the vector field $\mathbf{F}(x, y, z) = (z + y, x - z, x - y)$. Compute the flux out of the "closed" cone $z^2 = x^2 + y^2$ with $z \in [0, 1]$, capped off with a disc in the $z = 1$ plane in two ways: first directly using parameterizations, then by invoking the divergence theorem. Finally, verify that your answers agree.
2. Use the divergence theorem to compute the *inward* flux of the field

$$\mathbf{F}(x, y, z) = (2xz, 1 - 4xy^2, 2z - z^2)$$

into the surface S bound by the paraboloid $z = 1 - x^2 - y^2$ and the xy -plane.

Applications

3. Suppose that a velocity field for a fluid is given by $\mathbf{F}(x, y, z) = (x/y, x/z, -z/y)$. Compute the flow rate of the fluid across the cube $[-3, 4] \times [1, 2] \times [2, 8]$.

Note: Fluids with a velocity field $\mathbf{F}$ satisfying $\operatorname{div} \mathbf{F} = 0$ everywhere are called *incompressible*, since they are not allowed to "expand" or "shrink" at any point. Most liquids are only slightly compressible (i.e., $\operatorname{div} \mathbf{F} \approx 0$), so the theory of incompressible fluids is relatively robust for most applications.

Extensions

4. Consider the vector field $\mathbf{F}(x, y, z) = \left(\frac{1}{z} \cos(xz), y \sin(xz), 1\right)$. Compute the flux through the lateral walls (i.e., excluding the top and bottom) of the cylinder $x^2 + y^2 = 1$, with $1 \leq z \leq 2$. [Hint: This is a problem about a surface which isn't closed... why is it in the divergence theorem worksheet?]
5. Consider the radial vector field

$$\mathbf{F}(x, y, z) = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} \langle x, y, z \rangle$$

- (a) Consider the region between two concentric spheres: $V = \{1 \leq x^2 + y^2 + z^2 \leq R^2\} \subset \mathbb{R}^3$, where R is any number bigger than 1. Calculate the *net* flux across the boundary of this region.

- (b) Compute the outward flux of $\mathbf{F}$ over *any sphere* of radius R centered at the origin:
 $x^2 + y^2 + z^2 = R^2$.
- (c) Generalize your answer to the previous problem:

The outward flux of $\mathbf{F}$ over *any smooth surface* containing the origin equals
_____. [Note: this is essentially Gauss' Law for a point charge!]